

Program : Diploma in Communication and Computer Networking	
Course Code : 4326	Course Title: Network Analysis Lab
Semester: 4	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Troubleshooting using software utility/commands
- Troubleshooting using packet analysis Tool
- Troubleshooting using Hardware Tools.

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Perform troubleshooting using software utility/commands PO4-3, PO5-1, PO9-3, PO12-2	8	Applying
CO2	Perform packet Analysis using Wireshark and analysis of core applications PO4-3, PO1-3, PO5-2	10	Applying
CO3	Identify the basic hardware tools	9	Applying
CO4	Perform troubleshooting using hardware tools	10	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3		2		
CO2	3	3	3		2		
CO3	3		3		2		
CO4	3		3		2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Perform troubleshooting using software utility/commands		
M1.01	Familiarization of network trouble shooting commands - ping, traceroute/tracert, ipconfig/ifconfig, nslookup, netstat, putty/tera term, subnet and IP calculator, speedtest.net /pingtest.net ,pathping / mtr route , iptables, tcpdump, nmap, arp, dig	8	Applying
CO2	Perform packet Analysis using Wire shark		
M2.01	Familiarization of Packet Analysis Using Wire shark	3	Applying
M2.02	Perform packet analysis of ARP, IPv4 , IPv6 and ICMP	2	
M2.03	Analysing core service UDP ,DHCP ,DNS	2	
M2.04	Analysis of core network applications : FTP, HTTP, SSL ,Wi-Fi/802.11	3	
	Series Test– I	2	
CO3	Identify the basic hardware tools		
M3.01	Study of basic wire crimping mechanism	3	Applying
M3.02	Study of Cable testers - Testing of network cables	3	Applying
M3.03	Study of Punch Down tool : Connecting network cable directly to patch panel or puch down block)	3	Applying
CO4	Perform troubleshooting using hardware tools		
M4.01	Measuring of reflected signals using TDR/OTDR	2	Applying
M4.02	Measuring of power in an optical signal using Light meter	2	Applying
M4.03	Electrical signal checking using tone generator	2	Applying
M4.04	Trouble shooting network transmission issues using Loop back adapter	2	Applying

M4.05	Use of Multimeter and Spectrum Analyzer	2	Applying
	Test –II	2	
	Open Ended Experiments / micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.
Minimum of 6 experiments (excluding open ended experiments/micro projects) shall be performed.
Compulsory for CIA
The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text / Reference

T/R	Book Title/Author
R1	Practical computer network analysis and design by McCabe, James D
R2	Computer Networking: A Top-Down Approach Kurose, Ross
R3	Wireshark Network Analysis (Second Edition): The Official Wireshark Certified Network Analyst Study Guide by Laura Chappell (Author), Gerald Combs (Foreword)

Online Resources

Sl.No	Website Link
1	http://vlabs.iitkgp.ac.in/ant
2	https://ocean.sagepub.com/online-network-analysis-voson-lab